In [1]:
```r
library(Seurat)
# library(SeuratDisk)

library(reticulate)
library(anndata)

library(ggplot2)
library(ggpubr)
library(pheatmap)
library(dplyr)
library(tidyr)
library(RColorBrewer)
library(clustree)
library(repr)
options(repr.plot.width=10, repr.plot.height=8)

library(UpSetR)
library(grid)

library(PRROC)
library(Matrix)

# function to clean symbols from string
alphanumeric <- function(s) {
  gsub("[^[:alnum:]]", "", s)
}

getwd()

dataset_id <- "simulated_mm_RA"
genome_id <- "mm10"
level <- "family"
samples <- c("young", "old", "all")

for (sample in samples) {
    dir.create(paste0("figures_", dataset_id, "_family_", sample))
}
dir.create(paste0("figures_", dataset_id, "_family"))

thrMinCells <- 500 * 0.05
```

```
Loading required package: SeuratObject

Loading required package: sp

‘SeuratObject’ was built under R 4.3.3 but the current version is
4.4.1; it is recomended that you reinstall ‘SeuratObject’ as the ABI
for R may have changed

‘SeuratObject’ was built with package ‘Matrix’ 1.6.4 but the current
version is 1.7.5; it is recomended that you reinstall ‘SeuratObject’ as
the ABI for ‘Matrix’ may have changed


Attaching package: ‘SeuratObject’


The following objects are masked from ‘package:base’:

    intersect, t



Attaching package: ‘anndata’


The following object is masked from ‘package:SeuratObject’:

    Layers


Warning message:
“package ‘dplyr’ was built under R version 4.4.3”

Attaching package: ‘dplyr’


The following objects are masked from ‘package:stats’:

    filter, lag


The following objects are masked from ‘package:base’:

    intersect, setdiff, setequal, union


Warning message:
“package ‘tidyr’ was built under R version 4.4.3”
Warning message:
“package ‘RColorBrewer’ was built under R version 4.4.3”
Loading required package: ggraph


Attaching package: ‘ggraph’


The following object is masked from ‘package:sp’:

    geometry


Loading required package: rlang

Warning message:
“package ‘rlang’ was built under R version 4.4.3”

Attaching package: ‘Matrix’


The following objects are masked from ‘package:tidyr’:

    expand, pack, unpack

```

'/mnt/TEdata/TEbenchmarking/Rscripts'

```
Warning message in dir.create(paste0("figures_", dataset_id, "_family_", sample)):
“'figures_simulated_mm_RA_family_young' already exists”
Warning message in dir.create(paste0("figures_", dataset_id, "_family_", sample)):
“'figures_simulated_mm_RA_family_old' already exists”
Warning message in dir.create(paste0("figures_", dataset_id, "_family_", sample)):
“'figures_simulated_mm_RA_family_all' already exists”
Warning message in dir.create(paste0("figures_", dataset_id, "_family")):
“'figures_simulated_mm_RA_family' already exists”
```

In [2]:
```
# import workspace
load("workspaces/00_evaluation_objectCreation.Rdata")

colorTools <- c(
    "scTE" = "#E03756",
    "IRescue" = "#F88475",
    "STARsolo_TE" = "#FFBC81",
    "STARsolo_TE_EM" = "#F3E088",
    "SoloTE_unique" = "#A4DD9B",
    "SoloTE_thr2" = "#6FC69D",
    "SoloTE_thr1" = "#4BB2BB",
    "SoloTE_thr0" = "#3989BF",
    "Stellarscope" = "#4F5D93",
    "simulated" = "grey70"
)
```

Aggregate splatter matrices

In [4]:
```
# Aggregate splatter matrices by family
splatter_objs_family <- list()

for(sample in samples){
    families <- conversion_table$family[match(Features(splatter_objs[[sample]]), conversion_table$stellarscopeI

    print(table(conversion_table$class[match(Features(splatter_objs[[sample]]), conversion_table$stellarscopeID

    locus_counts <- GetAssayData(splatter_objs[[sample]], layer="counts")

    # Aggregate rows by family using the 'by' function
    family_counts <- by(locus_counts, families, colSums)
    # Convert the result to a proper matrix
    result_matrix <- do.call(rbind, family_counts)

    splatter_objs_family[[sample]] <- Seurat::CreateSeuratObject(result_matrix,
                                                                 project = splatter_objs[[sample
                                                                 min.cells = 0, min.features = 0

}
splatter_objs_family
```

```
 DNA LINE  LTR   RC SINE
1128  820 1693   93 1213
Warning message:
“Data is of class matrix. Coercing to dgCMatrix.”
 DNA LINE  LTR   RC SINE
1147  546  577   97  663
Warning message:
“Data is of class matrix. Coercing to dgCMatrix.”
LINE  LTR SINE
 286 1157  587
Warning message:
“Data is of class matrix. Coercing to dgCMatrix.”
$all
An object of class Seurat
32 features across 500 samples within 1 assay
Active assay: RNA (32 features, 0 variable features)
 1 layer present: counts

$old
An object of class Seurat
32 features across 500 samples within 1 assay
Active assay: RNA (32 features, 0 variable features)
 1 layer present: counts

$young
An object of class Seurat
7 features across 500 samples within 1 assay
Active assay: RNA (7 features, 0 variable features)
 1 layer present: counts
```

Aggregate tool matrices

In [5]:
```
objListfamily <- list()

for(tool in names(objList)){

  for(sample in samples){
```

```
    families <- conversion_table$family[match(Features(objList[[tool]][[sample]]), conversion_table$stellarscope

    locus_counts <- GetAssayData(objList[[tool]][[sample]], layer="counts")

    # Aggregate rows by family using the 'by' function
    family_counts <- by(locus_counts, families, colSums)
    # Convert the result to a proper matrix
    result_matrix <- do.call(rbind, family_counts)

    objListfamily[[tool]][[sample]] <- Seurat::CreateSeuratObject(result_matrix,
                                                                     project = objList[[tool]][[samp
                                                                     min.cells = thrMinCells, min.fea
}
}

names(objListfamily)
```

```
Warning message:
“Data is of class matrix. Coercing to dgCMatrix.”
Warning message:
“Data is of class matrix. Coercing to dgCMatrix.”
Warning message:
“Data is of class matrix. Coercing to dgCMatrix.”
Warning message:
“Data is of class matrix. Coercing to dgCMatrix.”
Warning message:
“Data is of class matrix. Coercing to dgCMatrix.”
Warning message:
“Data is of class matrix. Coercing to dgCMatrix.”
Warning message:
“Data is of class matrix. Coercing to dgCMatrix.”
Warning message:
“Data is of class matrix. Coercing to dgCMatrix.”
Warning message:
“Data is of class matrix. Coercing to dgCMatrix.”
Warning message:
“Data is of class matrix. Coercing to dgCMatrix.”
Warning message:
“Data is of class matrix. Coercing to dgCMatrix.”
Warning message:
“Data is of class matrix. Coercing to dgCMatrix.”
Warning message:
“Data is of class matrix. Coercing to dgCMatrix.”
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“Data is of class matrix. Coercing to dgCMatrix.”
Warning message:
“Data is of class matrix. Coercing to dgCMatrix.”
Warning message:
“Data is of class matrix. Coercing to dgCMatrix.”
Warning message:
“Data is of class matrix. Coercing to dgCMatrix.”
Warning message:
“Data is of class matrix. Coercing to dgCMatrix.”
Warning message:
“Data is of class matrix. Coercing to dgCMatrix.”
Warning message:
“Data is of class matrix. Coercing to dgCMatrix.”
```

'STARsolo_TE' · 'STARsolo_TE_EM' · 'SoloTE_unique' · 'SoloTE_thr2' · 'SoloTE_thr1' · 'SoloTE_thr0' · 'Stellarscope'

Import IRescue subfamily level matrix for young TEs and aggregate to family level

In [6]:

```
# add stripped subfamily and family IDs
conversion_table$subfamily_commonNames <- gsub(pattern="-", replacement="_", conversion_table$subfamily)
conversion_table$family_commonNames <- gsub(pattern="-", replacement="_", conversion_table$family)
```

In [7]:

```
sample <- "young"

main_path <- paste0("/mnt/TEdataStorage_2T/results/irescue/", dataset_id, "/", sample, "/counts/")

irescue_objs <- list()

# Read the three gzipped files
irescue_matrix  <- Seurat::Read10X(main_path)

print("Col sums:")
summary(colSums(irescue_matrix))
```

```
# # change - to _ to find common subfamily names
rownames(irescue_matrix) <- gsub(pattern="-", replacement="_", rownames(irescue_matrix))
print("Subfamilies in conversion table:")
print(table(rownames(irescue_matrix) %in% conversion_table$subfamily_commonNames))

print("Total counts of the subfamilies not present in our annotation are all 0:")
rowSums(irescue_matrix[rownames(irescue_matrix)[!(rownames(irescue_matrix) %in% conversion_table$subfamily_comm

# remove rows with 0 counts
irescue_matrix <- irescue_matrix[rowSums(irescue_matrix)>0,]

# match to names used for other tools
rownames(irescue_matrix) <- conversion_table$subfamily[match(rownames(irescue_matrix), conversion_table$subfami

print("Non-zero subfamilies in conversion table:")
table(rownames(irescue_matrix) %in% conversion_table$subfamily)

irescue_objs[[sample]] <- Seurat::CreateSeuratObject(irescue_matrix,
    project = paste0("IRescue_", sample),
    min.cells = thrMinCells, min.features = 0
)

families <- conversion_table$family[match(Features(irescue_objs[[sample]]), conversion_table$subfamily)]
subfamily_counts <- GetAssayData(irescue_objs[[sample]], layer="counts")

# Aggregate rows by family using the 'by' function
family_counts <- by(subfamily_counts, families, colSums)
# Convert the result to a proper matrix
result_matrix <- do.call(rbind, family_counts)

objListfamily[["IRescue"]][[sample]] <- Seurat::CreateSeuratObject(result_matrix,
        project = irescue_objs[[sample]]@project.name,
        min.cells = thrMinCells, min.features = 0)
objListfamily[["IRescue"]]
```

```
[1] "Col sums:"
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
  23439   37069   40927   41093   44896   65719 
[1] "Subfamilies in conversion table:"
FALSE  TRUE 
    8   921 
[1] "Total counts of the subfamilies not present in our annotation are all 0:"
```

LFSINE_Vert: 0 **MamSINE1:** 0 **U1:** 0 **U17:** 0 **U2:** 0 **U5:** 0 **U6:** 0 **U7:** 0

```
[1] "Non-zero subfamilies in conversion table:"
TRUE 
 212 
```

```
Warning message:
“Data is of class matrix. Coercing to dgCMatrix.”
$young
An object of class Seurat 
7 features across 500 samples within 1 assay 
Active assay: RNA (7 features, 0 variable features)
 1 layer present: counts
```

Import scTE sub-family level matrix for young TEs

In [8]:

```
sample <- "young"

main_path <- paste0("/mnt/TEdataStorage_2T/results/scTE/", sample, ".csv")

scTE_objs <- list()

# read matrix
scTE_matrix  <- read.csv(main_path, row.names=1)
scTE_matrix <- as.matrix(scTE_matrix)
scTE_matrix <- t(scTE_matrix)

print("Col sums:")
summary(colSums(scTE_matrix))

# remove rows with 0 counts
scTE_matrix <- scTE_matrix[rowSums(scTE_matrix)>0,]

summary(rowSums(scTE_matrix))
```

```
[1] "Col sums:"
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
  25564   38157   41618   41685   45116   63563 
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
      1       1       6    6881      36 2677306 
```

In [9]:
```r
# import annotations used by scTE
scTE_TEannotation <- read.table("/mnt/TEdata/TEbenchmarking/data/TEannotation/scTE/rmsk.txt.gz")
head(scTE_TEannotation)

scTE_gene_gtf <- read.table("/mnt/TEdata/TEbenchmarking/data/TEannotation/scTE/gencode.vM21.annotation.gtf.gz",
                  col.names=c("seqname","source","feature","start","end","score","strand","frame","attributes")
scTE_gene_gtf$gene_name <- gsub('.*gene_name ([^;]+);.*', '\\1', scTE_gene_gtf$attributes)

scTE_genes <- unique(scTE_gene_gtf$gene_name)
head(scTE_genes)
```

A data.frame: 6 × 17

	V1	V2	V3	V4	V5	V6	V7	V8	V9	V10	V11	V12	
	<int>	<int>	<int>	<int>	<int>	<chr>	<int>	<int>	<int>	<chr>	<chr>	<chr>	<
1	607	12955	105	9	10	chr1	3000000	3002128	-192469843	-	L1_Mus3	LINE	
2	607	1216	268	31	105	chr1	3003152	3003994	-192467977	-	L1Md_F	LINE	
3	607	234	279	0	0	chr1	3003993	3004054	-192467917	-	L1_Mus3	LINE	
4	607	3685	199	21	14	chr1	3004040	3004206	-192467765	+	L1_Rod	LINE	
5	607	376	62	31	0	chr1	3004206	3004270	-192467701	+	(CAAA)n	Simple_repeat	Simple_re
6	607	3685	199	21	14	chr1	3004270	3005001	-192466970	+	L1_Rod	LINE	

'4933401J01Rik' · 'Gm26206' · 'Xkr4' · 'Gm18956' · 'Gm37180' · 'Gm37363'

In [10]:
```r
# create df of scTE features and annotate them as genes and TEs

scTE_feature_df <- as.data.frame(rownames(scTE_matrix))
colnames(scTE_feature_df) <- "scTE_name"

scTE_feature_df$feature_type <- "unknown"

scTE_feature_df$feature_type[alphanumeric(scTE_feature_df$scTE_name) %in% alphanumeric(scTE_TEannotation$V11)] <
scTE_feature_df$feature_type[alphanumeric(scTE_feature_df$scTE_name) %in% alphanumeric(scTE_genes)] <- "gene"
scTE_feature_df$feature_type[grepl("Rik",scTE_feature_df$scTE_name)] <- "gene"

table(scTE_feature_df$feature_type)
```

```
gene   TE
2454  575
```

In [11]:
```r
# Check how many detected genes are expressed in more than 5% of cells
scTE_matrix_genes <- scTE_matrix[scTE_feature_df$scTE_name[scTE_feature_df$feature_type=="gene"],]
dim(scTE_matrix_genes[rowSums(scTE_matrix_genes) > thrMinCells,])
```

615 · 500

In [12]:
```r
# keep only TEs in the scTE matrix
scTE_matrix <- scTE_matrix[scTE_feature_df$scTE_name[scTE_feature_df$feature_type=="TE"],]
dim(scTE_matrix)
```

575 · 500

In [13]:
```r
# change - and . to _ to find common subfamily names
rownames(scTE_matrix) <- gsub(pattern="-", replacement="_", rownames(scTE_matrix))
rownames(scTE_matrix) <- gsub(pattern="\\.", replacement="_", rownames(scTE_matrix))


print("Subfamily not present in our annotation:")
rownames(scTE_matrix)[!(rownames(scTE_matrix) %in% conversion_table$subfamily_commonNames)] # not in our convers

# match to names used for other tools (for those that are matched)
matched <- conversion_table$subfamily[match(rownames(scTE_matrix), conversion_table$subfamily_commonNames)]
rownames(scTE_matrix) <- ifelse(is.na(matched), rownames(scTE_matrix), matched)

print("Non-zero subfamilies in conversion table:")
table(rownames(scTE_matrix) %in% conversion_table$subfamily)

scTE_objs[[sample]] <- Seurat::CreateSeuratObject(scTE_matrix,
    project = paste0("scTE_", sample),
    min.cells = thrMinCells, min.features = 0
)
scTE_objs

families <- conversion_table$family[match(Features(scTE_objs[[sample]]), conversion_table$subfamily)]
subfamily_counts <- GetAssayData(scTE_objs[[sample]], layer="counts")

# Aggregate rows by family using the 'by' function
```

```
family_counts <- by(subfamily_counts, families, colSums)
# Convert the result to a proper matrix
result_matrix <- do.call(rbind, family_counts)

objListfamily[["scTE"]][[sample]] <- Seurat::CreateSeuratObject(result_matrix,
        project = scTE_objs[[sample]]@project.name,
        min.cells = thrMinCells, min.features = 0)
objListfamily[["scTE"]]
```

```
[1] "Subfamily not present in our annotation:"
```

'LFSINE_Vert'

```
[1] "Non-zero subfamilies in conversion table:"
FALSE  TRUE
    1   574
```

```
Warning message:
“Feature names cannot have underscores ('_'), replacing with dashes ('-')”
Warning message:
“Data is of class matrix. Coercing to dgCMatrix.”
```

```
$young
An object of class Seurat
260 features across 500 samples within 1 assay
Active assay: RNA (260 features, 0 variable features)
 1 layer present: counts
```

```
Warning message:
“Data is of class matrix. Coercing to dgCMatrix.”
```

```
$young
An object of class Seurat
14 features across 500 samples within 1 assay
Active assay: RNA (14 features, 0 variable features)
 1 layer present: counts
```

Statistics

In [14]:

```
for(sample in samples){
  print(sample)
  print(splatter_objs_family[[sample]]@project.name)
  print(sum(splatter_objs_family[[sample]]$nCount_RNA))
  nCounts_sample <- vector()

  for(tool in names(objListfamily)){
    if(sample %in% names(objListfamily[[tool]])){
      obj <- objListfamily[[tool]][[sample]]
      nCounts_sample[obj@project.name] <- sum(obj$nCount_RNA)
    }
  }
  print(nCounts_sample)
}
```

```
[1] "all"
[1] "simulated_all"
[1] 30376911
   STARsolo_TE_all STARsolo_TE_EM_all  SoloTE_unique_all    SoloTE_thr2_all
          20864548           28093185           19365845           20256586
   SoloTE_thr1_all    SoloTE_thr0_all   Stellarscope_all
          21087351           28829577           26733978
[1] "old"
[1] "simulated_old"
[1] 30384744
   STARsolo_TE_old STARsolo_TE_EM_old  SoloTE_unique_old    SoloTE_thr2_old
          29869669           30342764           31605699           31841727
   SoloTE_thr1_old    SoloTE_thr0_old   Stellarscope_old
          31924535           32271772           31666805
[1] "young"
[1] "simulated_young"
[1] 30392717
   STARsolo_TE_young STARsolo_TE_EM_young  SoloTE_unique_young
             9424570             25014811              3805194
   SoloTE_thr2_young    SoloTE_thr1_young    SoloTE_thr0_young
             6337946              8279167             25812726
  Stellarscope_young        IRescue_young           scTE_young
            22135230             20545865             20415903
```

In [15]:

```
## n TPs
for(sample in samples){
  print(sample)
  print("Simulated")
  print(length(Features(splatter_objs_family[[sample]])))

  for(tool in names(objListfamily)){
    if(sample %in% names(objListfamily[[tool]])){
      obj <- objListfamily[[tool]][[sample]]
```

```r
        print(obj@project.name)
        print(length(intersect(Features(splatter_objs_family[[sample]]), Features(obj))))
      }
    }
  }
```

```
[1] "all"
[1] "Simulated"
[1] 32
[1] "STARsolo_TE_all"
[1] 32
[1] "STARsolo_TE_EM_all"
[1] 32
[1] "SoloTE_unique_all"
[1] 32
[1] "SoloTE_thr2_all"
[1] 32
[1] "SoloTE_thr1_all"
[1] 32
[1] "SoloTE_thr0_all"
[1] 32
[1] "Stellarscope_all"
[1] 32
[1] "old"
[1] "Simulated"
[1] 32
[1] "STARsolo_TE_old"
[1] 32
[1] "STARsolo_TE_EM_old"
[1] 32
[1] "SoloTE_unique_old"
[1] 32
[1] "SoloTE_thr2_old"
[1] 32
[1] "SoloTE_thr1_old"
[1] 32
[1] "SoloTE_thr0_old"
[1] 32
[1] "Stellarscope_old"
[1] 32
[1] "young"
[1] "Simulated"
[1] 7
[1] "STARsolo_TE_young"
[1] 7
[1] "STARsolo_TE_EM_young"
[1] 7
[1] "SoloTE_unique_young"
[1] 7
[1] "SoloTE_thr2_young"
[1] 7
[1] "SoloTE_thr1_young"
[1] 7
[1] "SoloTE_thr0_young"
[1] 7
[1] "Stellarscope_young"
[1] 7
[1] "IRescue_young"
[1] 7
[1] "scTE_young"
[1] 7
```

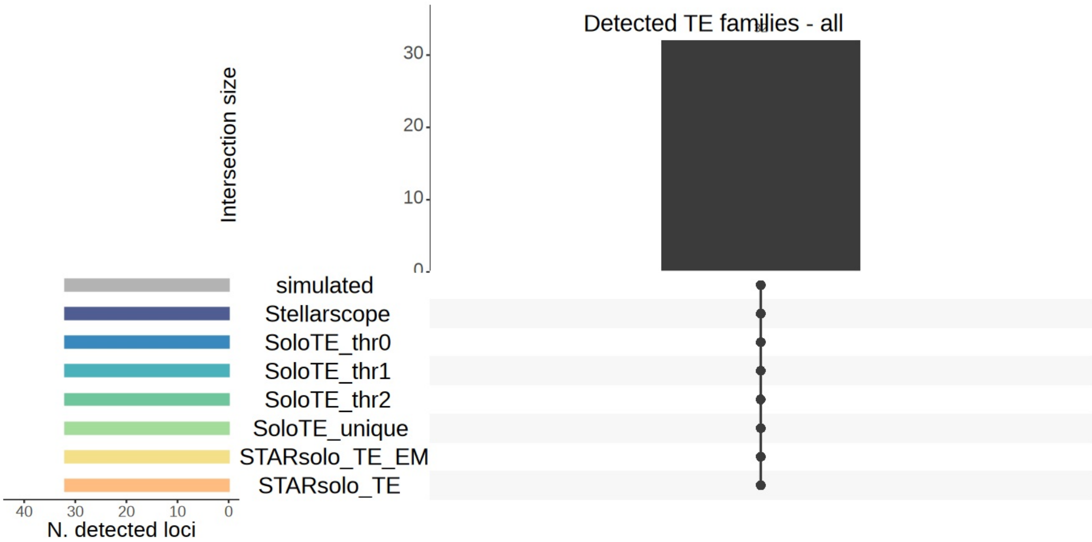
In [16]:

```r
options(repr.plot.width=12, repr.plot.height=6)


for (sample in samples){

    print(sample)


    TPlist <- list()
    detectedList <- list()

    # get tools that were run for the current sample and reorder based on color
    tools_with_sample <- names(Filter(function(tool) sample %in% names(tool), objListfamily))
    tools_with_sample <- names(colorTools)[names(colorTools) %in% tools_with_sample]


    for (tool in c(tools_with_sample, "simulated")){

        print(tool)
        if (tool == "simulated") {
            obj <- splatter_objs_family[[sample]]
        }else {
```

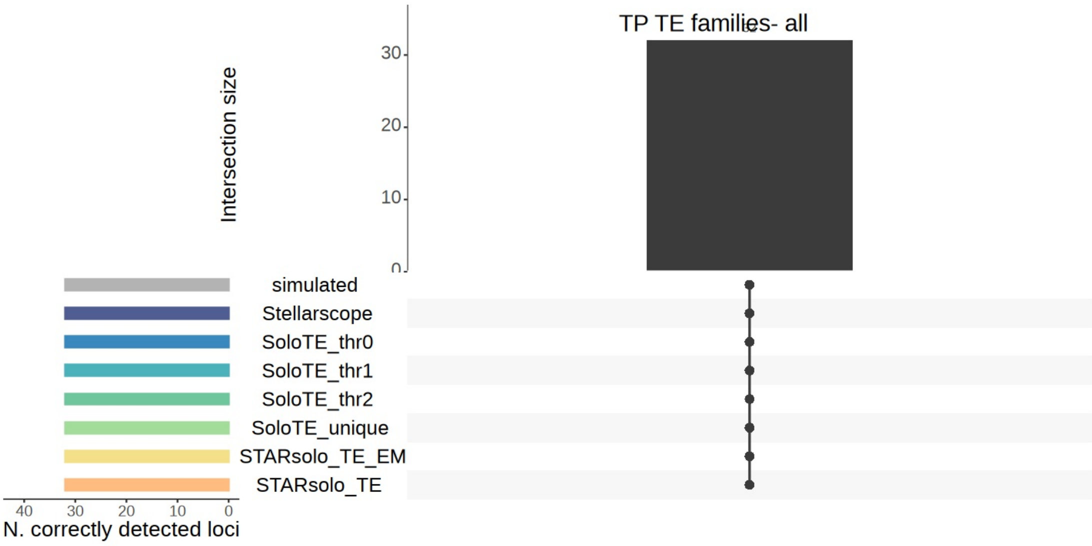
```r
            obj <- objListfamily[[tool]][[sample]]
        }
        print(obj)
        detectedList[[sub("^(.*)_.*$", "\\1", obj@project.name)]] <- Features(obj)
        TPlist[[sub("^(.*)_.*$", "\\1", obj@project.name)]] <- intersect(Features(splatter_objs_family[[sample]]
    }

    #pdf(file=paste0("figures/figures_simulated_mm_RA_family_",sample,"/upset_detected.pdf"), width = 13, heigh
    show(UpSetR::upset(fromList(detectedList), nintersects = 30,
            sets=c(tools_with_sample, "simulated"), keep.order = T, sets.bar.color=colorTools[c(tools_with_samp
            text.scale = c(2, 2, 2, 1.65, 2.5, 1.5),
            order.by=c("freq"),
            point.size=3, mb.ratio = c(0.5, 0.5),
            set_size.show=F, set_size.scale_max= max(lengths(detectedList))+10, #show.numbers = F,
            nsets=length(c(tools_with_sample, "simulated")),
            sets.x.label="N. detected loci",
            mainbar.y.label="Intersection size")

    )
    grid.text(paste0("Detected TE families - ", sample ),x = 0.65, y=0.95, gp=gpar(fontsize=18))
    #dev.off()

    #pdf(file=paste0("figures/figures_simulated_mm_RA_subfamily_",sample,"/upset_detected_TP.pdf"), width = 11,
    show(UpSetR::upset(fromList(TPlist), nintersects = 20,
            sets=c(tools_with_sample, "simulated"), keep.order = T, sets.bar.color=colorTools[c(tools_with_samp
            text.scale = c(2, 2, 2, 1.65, 2.2, 1.5),
            order.by=c("freq"),
            point.size=3, mb.ratio = c(0.5, 0.5),
            set_size.show=F, set_size.scale_max= max(lengths(TPlist))+10, #show.numbers = F,
            nsets=length(c(tools_with_sample, "simulated")),
            sets.x.label="N. correctly detected loci",
            mainbar.y.label="Intersection size")
    )
    grid.text(paste0("TP TE families- ", sample ),x = 0.65, y=0.95, gp=gpar(fontsize=18))
    #dev.off()
}
```

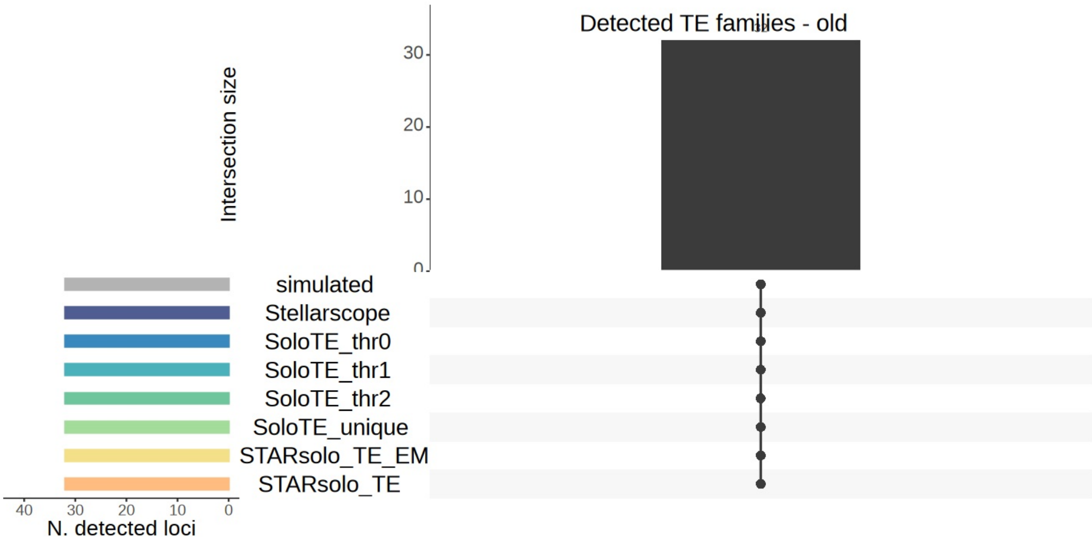
```
[1] "all"
[1] "STARsolo_TE"
An object of class Seurat
32 features across 500 samples within 1 assay
Active assay: RNA (32 features, 0 variable features)
 1 layer present: counts
[1] "STARsolo_TE_EM"
An object of class Seurat
32 features across 500 samples within 1 assay
Active assay: RNA (32 features, 0 variable features)
 1 layer present: counts
[1] "SoloTE_unique"
An object of class Seurat
32 features across 500 samples within 1 assay
Active assay: RNA (32 features, 0 variable features)
 1 layer present: counts
[1] "SoloTE_thr2"
An object of class Seurat
32 features across 500 samples within 1 assay
Active assay: RNA (32 features, 0 variable features)
 1 layer present: counts
[1] "SoloTE_thr1"
An object of class Seurat
32 features across 500 samples within 1 assay
Active assay: RNA (32 features, 0 variable features)
 1 layer present: counts
[1] "SoloTE_thr0"
An object of class Seurat
32 features across 500 samples within 1 assay
Active assay: RNA (32 features, 0 variable features)
 1 layer present: counts
[1] "Stellarscope"
An object of class Seurat
32 features across 500 samples within 1 assay
Active assay: RNA (32 features, 0 variable features)
 1 layer present: counts
[1] "simulated"
An object of class Seurat
32 features across 500 samples within 1 assay
Active assay: RNA (32 features, 0 variable features)
 1 layer present: counts
```

```
Warning message:
“`aes_string()` was deprecated in ggplot2 3.0.0.
ℹ Please use tidy evaluation idioms with `aes()`.
ℹ See also `vignette("ggplot2-in-packages")` for more information.
ℹ The deprecated feature was likely used in the UpSetR package.

  Please report the issue to the authors.”
Warning message:
“Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
ℹ Please use `linewidth` instead.
ℹ The deprecated feature was likely used in the UpSetR package.
  Please report the issue to the authors.”
Warning message:
“The `size` argument of `element_line()` is deprecated as of ggplot2 3.4.0.
ℹ Please use the `linewidth` argument instead.
ℹ The deprecated feature was likely used in the UpSetR package.
  Please report the issue to the authors.”
```

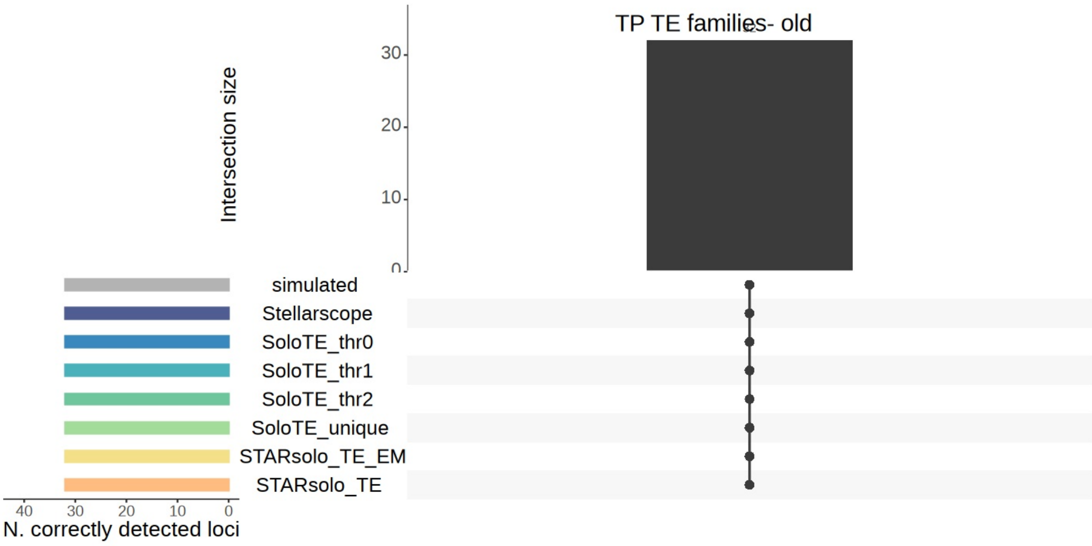


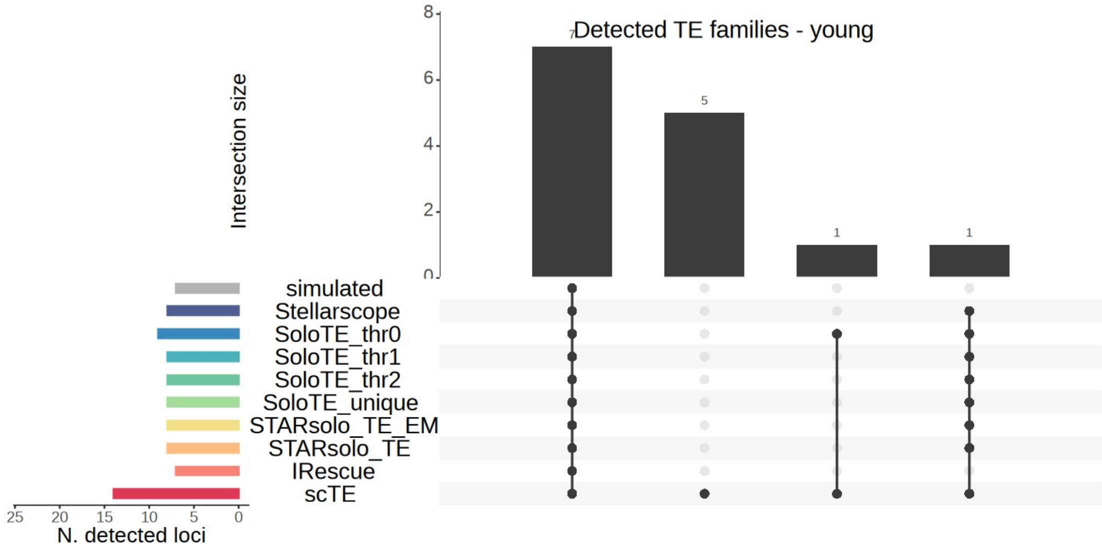

```
[1] "old"
[1] "STARsolo_TE"
An object of class Seurat
32 features across 500 samples within 1 assay
Active assay: RNA (32 features, 0 variable features)
 1 layer present: counts
[1] "STARsolo_TE_EM"
An object of class Seurat
32 features across 500 samples within 1 assay
Active assay: RNA (32 features, 0 variable features)
 1 layer present: counts
[1] "SoloTE_unique"
An object of class Seurat
32 features across 500 samples within 1 assay
Active assay: RNA (32 features, 0 variable features)
 1 layer present: counts
[1] "SoloTE_thr2"
An object of class Seurat
32 features across 500 samples within 1 assay
Active assay: RNA (32 features, 0 variable features)
 1 layer present: counts
[1] "SoloTE_thr1"
An object of class Seurat
32 features across 500 samples within 1 assay
Active assay: RNA (32 features, 0 variable features)
 1 layer present: counts
[1] "SoloTE_thr0"
An object of class Seurat
32 features across 500 samples within 1 assay
Active assay: RNA (32 features, 0 variable features)
 1 layer present: counts
[1] "Stellarscope"
An object of class Seurat
32 features across 500 samples within 1 assay
Active assay: RNA (32 features, 0 variable features)
 1 layer present: counts
[1] "simulated"
An object of class Seurat
32 features across 500 samples within 1 assay
Active assay: RNA (32 features, 0 variable features)
 1 layer present: counts
```

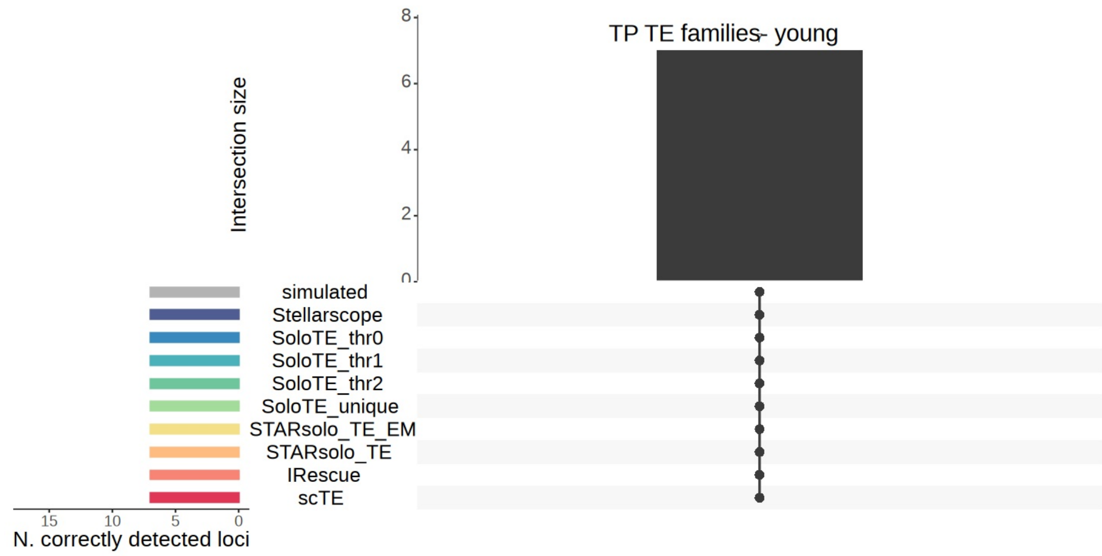

TP TE families- all
Intersection size
30
20
10
0
simulated
Stellarscope
SoloTE_thr0
SoloTE_thr1
SoloTE_thr2
SoloTE_unique
STARsolo_TE_EM
STARsolo_TE
40
30
20
10
0
N. correctly detected loci


Detected TE families - old
Intersection size
30
20
10
0
simulated
Stellarscope
SoloTE_thr0
SoloTE_thr1
SoloTE_thr2
SoloTE_unique
STARsolo_TE_EM
STARsolo_TE
40
30
20
10
0
N. detected loci

```
[1] "young"
[1] "scTE"
An object of class Seurat
14 features across 500 samples within 1 assay
Active assay: RNA (14 features, 0 variable features)
 1 layer present: counts
[1] "IRescue"
An object of class Seurat
7 features across 500 samples within 1 assay
Active assay: RNA (7 features, 0 variable features)
 1 layer present: counts
[1] "STARsolo_TE"
An object of class Seurat
8 features across 500 samples within 1 assay
Active assay: RNA (8 features, 0 variable features)
 1 layer present: counts
[1] "STARsolo_TE_EM"
An object of class Seurat
8 features across 500 samples within 1 assay
Active assay: RNA (8 features, 0 variable features)
 1 layer present: counts
[1] "SoloTE_unique"
An object of class Seurat
8 features across 500 samples within 1 assay
Active assay: RNA (8 features, 0 variable features)
 1 layer present: counts
[1] "SoloTE_thr2"
An object of class Seurat
8 features across 500 samples within 1 assay
Active assay: RNA (8 features, 0 variable features)
 1 layer present: counts
[1] "SoloTE_thr1"
An object of class Seurat
8 features across 500 samples within 1 assay
Active assay: RNA (8 features, 0 variable features)
 1 layer present: counts

[1] "SoloTE_thr0"
An object of class Seurat
9 features across 500 samples within 1 assay
Active assay: RNA (9 features, 0 variable features)
 1 layer present: counts
[1] "Stellarscope"
An object of class Seurat
8 features across 500 samples within 1 assay
Active assay: RNA (8 features, 0 variable features)
 1 layer present: counts
[1] "simulated"
An object of class Seurat
7 features across 500 samples within 1 assay
Active assay: RNA (7 features, 0 variable features)
 1 layer present: counts
```







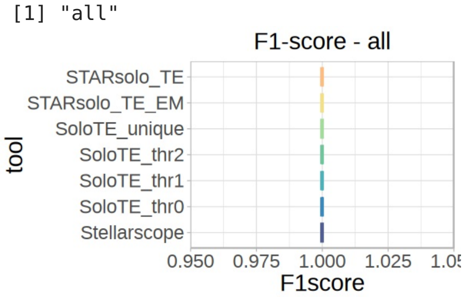
In [18]:

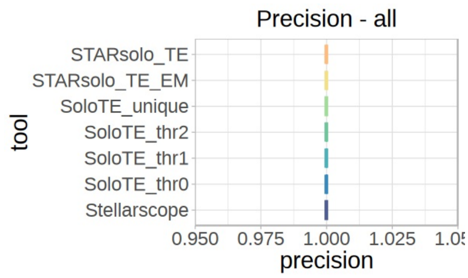
```
precisionTool <- list()
recallTool <- list()

for (sample in samples){
    layer <- "counts"

    matSplatter <- GetAssayData(splatter_objs_family[[sample]], layer=layer)

    precisionTool[[sample]] <- list()
    recallTool[[sample]] <- list()

    tools_with_sample <- names(Filter(function(tool) sample %in% names(tool), objListfamily))

    for(tool in tools_with_sample){

        matTool <- GetAssayData(objListfamily[[tool]][[sample]], layer=layer)

        precisionTool[[sample]][[tool]] <- NULL
        recallTool[[sample]][[tool]] <- NULL

        for(cell in Cells(splatter_objs_family[[sample]])){

            exprSplatter <- names(matSplatter[,cell])[matSplatter[,cell] > 0]
            # TEs expressed in the simulated matrix

            exprTool <- names(matTool[,cell])[matTool[,cell] > 0]
            # TEs detected by the tool

            TP <- intersect(exprSplatter, exprTool)
```

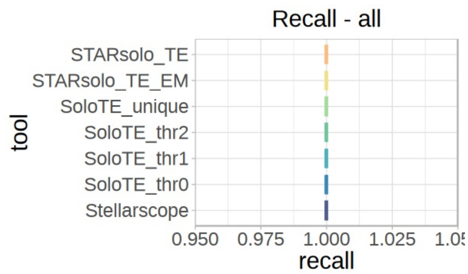
```
            nTP <- length(TP)

            FP <- setdiff(exprTool, exprSplatter)
            nFP <- length(FP)

            # add FP genes to the FP count
            if(tool %in% names(objGeneList)){
                FP_genes <- names(matTool_genes[,cell])[matTool_genes[,cell] > 0]
                nFP_genes <- length(FP_genes)
                nFP <- nFP + nFP_genes
            }

            FN <- setdiff(exprSplatter, exprTool)
            nFN <- length(FN)

            precision <- nTP / (nTP + nFP)
            precisionTool[[sample]][[tool]] <- c(precisionTool[[sample]][[tool]], precision)

            recall <- nTP / (nTP + nFN)
            recallTool[[sample]][[tool]] <- c(recallTool[[sample]][[tool]], recall)
        }
    }
}
```

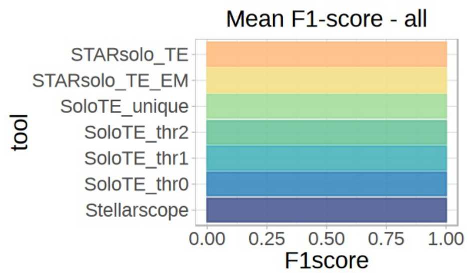
In [19]:
```
options(repr.plot.width=5, repr.plot.height=3)

for (sample in samples){
    print(sample)

    precisionDf <- stack(precisionTool[[sample]])
    colnames(precisionDf) <- c("precision", "tool")

    recallDf <- stack(recallTool[[sample]])
    colnames(recallDf) <- c("recall", "tool")

    df <- merge(precisionDf, recallDf, by='tool')

    df$F1score <- (2 * df$precision * df$recall) / (df$precision + df$recall)

    df$tool <- factor(df$tool, levels=rev(names(colorTools)))




    # plot just the mean
    precisionDf <- stack(lapply(precisionTool[[sample]], mean))
    colnames(precisionDf) <- c("precision", "tool")

    recallDf <- stack(lapply(recallTool[[sample]], mean))
    colnames(recallDf) <- c("recall", "tool")

    Fscores <- sapply(names(precisionTool[[sample]]), function(tool){
                        (2 * precisionTool[[sample]][[tool]] * recallTool[[sample]][[tool]]) /
                        (precisionTool[[sample]][[tool]] + recallTool[[sample]][[tool]])
    })
    meanFscores <- colMeans(Fscores)
    FscoreDf <- stack(meanFscores)
    colnames(FscoreDf) <- c("F1score", "tool")

    df <- merge(precisionDf, recallDf, by=c("tool"))
    df <- merge(df, FscoreDf, by=c("tool"))

    df$tool <- factor(df$tool, levels=rev(names(colorTools)))


}
```

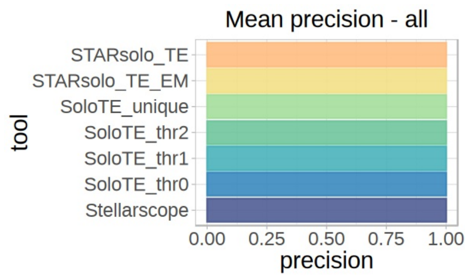
```
[1] "all"
```



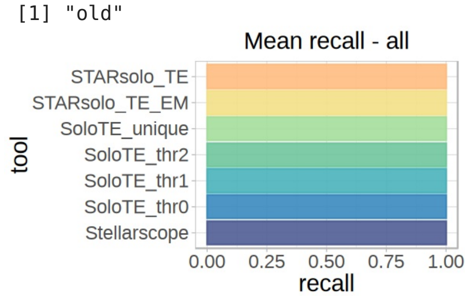

Precision - all
tool
STARsolo_TE
STARsolo_TE_EM
SoloTE_unique
SoloTE_thr2
SoloTE_thr1
SoloTE_thr0
Stellarscope
0.950 0.975 1.000 1.025
precision

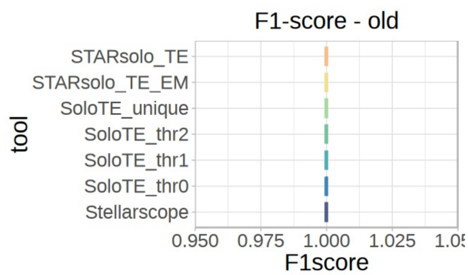

Recall - all
tool
STARsolo_TE
STARsolo_TE_EM
SoloTE_unique
SoloTE_thr2
SoloTE_thr1
SoloTE_thr0
Stellarscope
0.950 0.975 1.000 1.025
recall

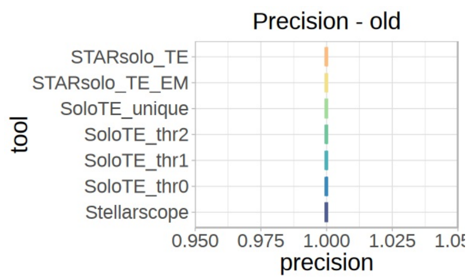

Mean F1-score - all
tool
STARsolo_TE
STARsolo_TE_EM
SoloTE_unique
SoloTE_thr2
SoloTE_thr1
SoloTE_thr0
Stellarscope
0.00 0.25 0.50 0.75 1.00
F1score

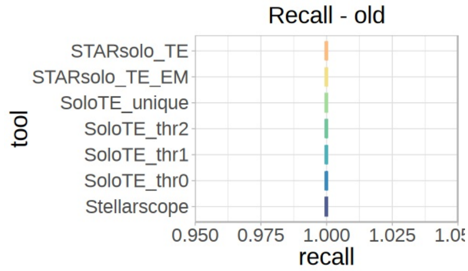

Mean precision - all
tool
STARsolo_TE
STARsolo_TE_EM
SoloTE_unique
SoloTE_thr2
SoloTE_thr1
SoloTE_thr0
Stellarscope
0.00 0.25 0.50 0.75 1.00
precision

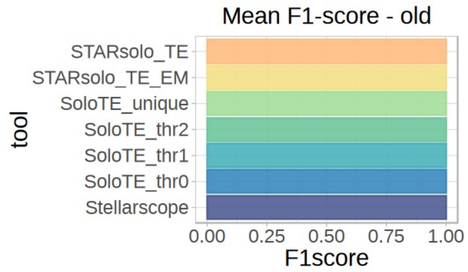
```
[1] "old"
```

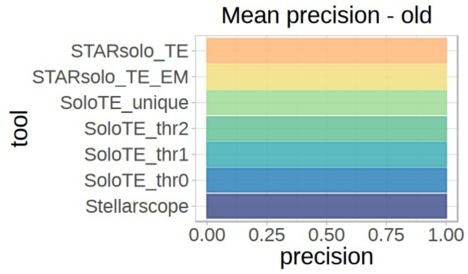

Mean recall - all
tool
STARsolo_TE
STARsolo_TE_EM
SoloTE_unique
SoloTE_thr2
SoloTE_thr1
SoloTE_thr0
Stellarscope
0.00 0.25 0.50 0.75 1.00
recall


F1-score - old
tool
STARsolo_TE
STARsolo_TE_EM
SoloTE_unique
SoloTE_thr2
SoloTE_thr1
SoloTE_thr0
Stellarscope
0.950 0.975 1.000 1.025
F1score

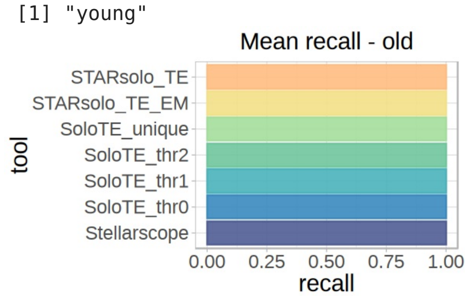

Precision - old
tool
STARsolo_TE
STARsolo_TE_EM
SoloTE_unique
SoloTE_thr2
SoloTE_thr1
SoloTE_thr0
Stellarscope
0.950 0.975 1.000 1.025
precision

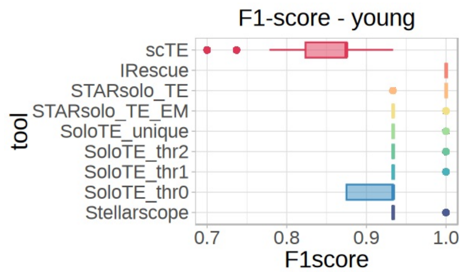

Recall - old
STARsolo_TE
STARsolo_TE_EM
SoloTE_unique
SoloTE_thr2
SoloTE_thr1
SoloTE_thr0
Stellarscope
tool
0.950 0.975 1.000 1.025
recall

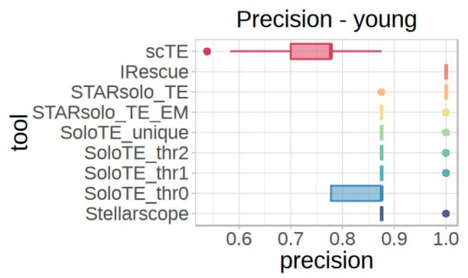

Mean F1-score - old
STARsolo_TE
STARsolo_TE_EM
SoloTE_unique
SoloTE_thr2
SoloTE_thr1
SoloTE_thr0
Stellarscope
tool
0.00 0.25 0.50 0.75 1.00
F1score

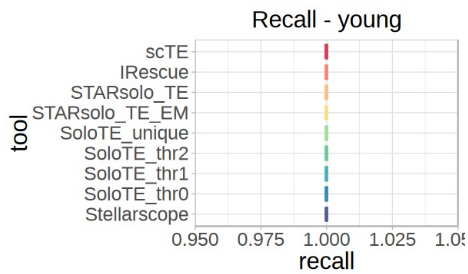

Mean precision - old
STARsolo_TE
STARsolo_TE_EM
SoloTE_unique
SoloTE_thr2
SoloTE_thr1
SoloTE_thr0
Stellarscope
tool
0.00 0.25 0.50 0.75 1.00
precision

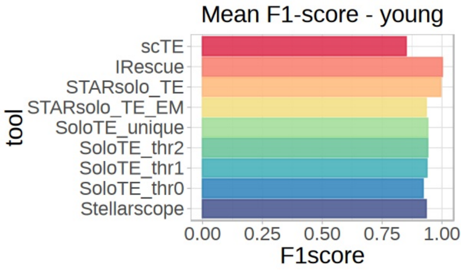
```
[1] "young"
```

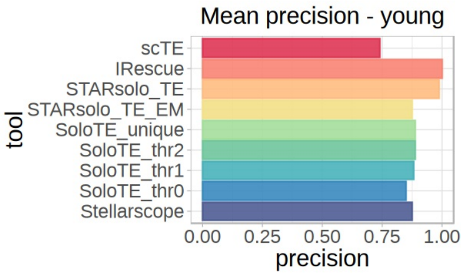

Mean recall - old
STARsolo_TE
STARsolo_TE_EM
SoloTE_unique
SoloTE_thr2
SoloTE_thr1
SoloTE_thr0
Stellarscope
tool
0.00 0.25 0.50 0.75 1.00
recall

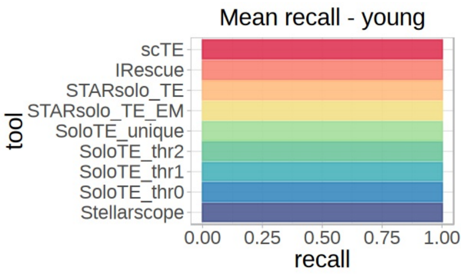

F1-score - young
scTE
IRescue
STARsolo_TE
STARsolo_TE_EM
SoloTE_unique
SoloTE_thr2
SoloTE_thr1
SoloTE_thr0
Stellarscope
tool
0.7 0.8 0.9 1.0
F1score


Precision - young
scTE
IRescue
STARsolo_TE
STARsolo_TE_EM
SoloTE_unique
SoloTE_thr2
SoloTE_thr1
SoloTE_thr0
Stellarscope
tool
0.6 0.7 0.8 0.9 1.0
precision


Recall - young
scTE
IRescue
STARsolo_TE
STARsolo_TE_EM
SoloTE_unique
SoloTE_thr2
SoloTE_thr1
SoloTE_thr0
Stellarscope
tool
0.950 0.975 1.000 1.025
recall







In [20]:

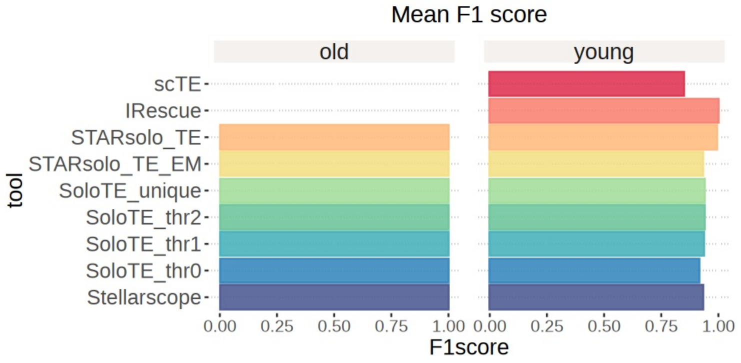
```
meanF1score_age_df <- NULL

for(age in c("old", "young")){

  # plot just the mean value
  precisionDf <- stack(lapply(precisionTool[[age]], mean))
  colnames(precisionDf) <- c("precision", "tool")

  recallDf <- stack(lapply(recallTool[[age]], mean))
  colnames(recallDf) <- c("recall", "tool")

  Fscores <- sapply(names(precisionTool[[age]]), function(tool){
                         (2 * precisionTool[[age]][[tool]] * recallTool[[age]][[tool]]) /
                         (precisionTool[[age]][[tool]] + recallTool[[age]][[tool]])
  })

  meanFscores <- colMeans(Fscores)
  FscoreDf <- stack(meanFscores)
  colnames(FscoreDf) <- c("F1score", "tool")

  df <- merge(precisionDf, recallDf, by=c("tool"))
  df <- merge(df, FscoreDf, by=c("tool"))

  df$tool <- factor(df$tool, levels=rev(names(colorTools)))

  # create df for facet plot
  meanF1score_age_df <- rbind(meanF1score_age_df, cbind(df, age))

}
```

In [22]:

```
options(repr.plot.width=8, repr.plot.height=4)

ggplot(meanF1score_age_df, aes(x=F1score, y=tool, color=tool, fill=tool)) +
    facet_wrap(~age, ncol = 2) +
    geom_col(alpha = 0.9) +
    scale_color_manual(values=colorTools) +
    scale_fill_manual(values=colorTools) +
    theme_pubclean() + ggtitle(paste0("Mean F1 score")) +
    guides(fill="none", color="none") +
  xlim(c(0,1)) +
    theme(text=element_text(size=17),
      plot.title = element_text(size=18, hjust=0.5),
      plot.subtitle = element_text(size=18, hjust=0.5),
      axis.text.y = element_text(size=16),
      axis.text.x = element_text(size=13),
```

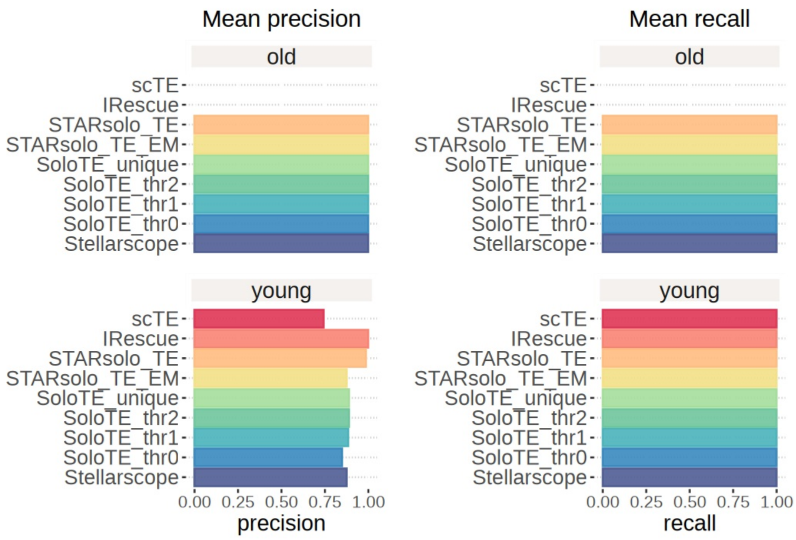
```
    strip.text = element_text(size=17, hjust=0.5, vjust = 0.5),
    strip.background = element_rect(fill="#F4F1EE", linewidth = 3, color = "white"),
    panel.spacing = unit(1, "lines"))
ggsave(paste0("figures/figures_simulated_mm_RA_family/meanF1_tool_byAge_horiz.png"), create.dir=TRUE)
ggsave(paste0("figures/figures_simulated_mm_RA_family/meanF1_tool_byAge_horiz.pdf"), device="pdf", height=4, wi
```

```
✔ Created directory: figures/figures_simulated_mm_RA_family.

Saving 6.67 x 6.67 in image
```

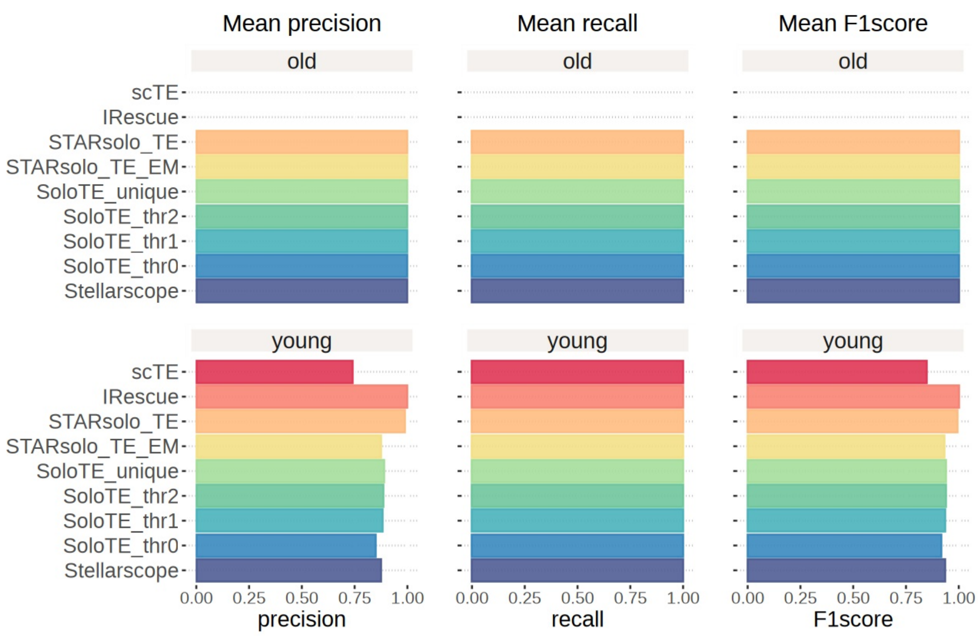


In [23]:
```
options(repr.plot.width=9, repr.plot.height=6)
colorTools <- c(
    "scTE" = "#E03756",
    "IRescue" = "#F88475",
    "STARsolo_TE" = "#FFBC81",
    "STARsolo_TE_EM" = "#F3E088",
    "SoloTE_unique" = "#A4DD9B",
    "SoloTE_thr2" = "#6FC69D",
    "SoloTE_thr1" = "#4BB2BB",
    "SoloTE_thr0" = "#3989BF",
    "Stellarscope" = "#4F5D93",
    "simulated" = "grey70"
)

precision_plot <- ggplot(meanF1score_age_df, aes(x=precision, y=tool, color=tool, fill=tool)) +
    facet_wrap(~age, ncol = 1) +
    geom_col(alpha = 0.9) +
    scale_color_manual(values=colorTools) +
    scale_fill_manual(values=colorTools) +
    theme_pubclean() + ggtitle(paste0("Mean precision")) +
    guides(fill="none", color="none") +
    xlim(c(0,1)) +
    ylab("") +
    theme(text=element_text(size=17),
      plot.title = element_text(size=18, hjust=0.5),
      plot.subtitle = element_text(size=18, hjust=0.5),
      axis.text.y = element_text(size=16),
      axis.text.x = element_text(size=13),
      strip.text = element_text(size=17, hjust=0.5, vjust = 0.5),
      strip.background = element_rect(fill="#F4F1EE", linewidth = 3, color = "white"),
      panel.spacing = unit(1, "lines"))

recall_plot <- ggplot(meanF1score_age_df, aes(x=recall, y=tool, color=tool, fill=tool)) +
    facet_wrap(~age, ncol = 1) +
    geom_col(alpha = 0.9) +
    scale_color_manual(values=colorTools) +
    scale_fill_manual(values=colorTools) +
    theme_pubclean() + ggtitle(paste0("Mean recall")) +
    guides(fill="none", color="none") +
    xlim(c(0,1)) +
    ylab("") +
    theme(text=element_text(size=17),
      plot.title = element_text(size=18, hjust=0.5),
      plot.subtitle = element_text(size=18, hjust=0.5),
      axis.text.y = element_text(size=16),
      axis.text.x = element_text(size=13),
      strip.text = element_text(size=17, hjust=0.5, vjust = 0.5),
      strip.background = element_rect(fill="#F4F1EE", linewidth = 3, color = "white"),
      panel.spacing = unit(1, "lines"))

f1score_plot <- ggplot(meanF1score_age_df, aes(x=F1score, y=tool, color=tool, fill=tool)) +
    facet_wrap(~age, ncol = 1) +
    geom_col(alpha = 0.9) +
    scale_color_manual(values=colorTools) +
    scale_fill_manual(values=colorTools) +
    theme_pubclean() + ggtitle(paste0("Mean F1score")) +
    guides(fill="none", color="none") +
    xlim(c(0,1)) +
    ylab("") +
```

```
    theme(text=element_text(size=17),
      plot.title = element_text(size=18, hjust=0.5),
      plot.subtitle = element_text(size=18, hjust=0.5),
      axis.text.y = element_text(size=16),
      axis.text.x = element_text(size=13),
      strip.text = element_text(size=17, hjust=0.5, vjust = 0.5),
      strip.background = element_rect(fill="#F4F1EE", linewidth = 3, color = "white"),
      panel.spacing = unit(1, "lines"))

precision_plot + recall_plot
ggsave(paste0("figures/figures_simulated_mm_RA_family/meanPrecisionRecall_tool_byAge.png"),
      height=6, width=9)

ggsave(paste0("figures/figures_simulated_mm_RA_family/meanPrecisionRecall_tool_byAge.pdf"),
      device="pdf", height=6, width=9)
```



In [25]:
```
options(repr.plot.width=11, repr.plot.height=7)

library(patchwork)

recall_plot_noyname <- recall_plot + theme(axis.text.y = element_blank())
f1_plot_noyname <- f1score_plot + theme(axis.text.y = element_blank())

(precision_plot + recall_plot_noyname + f1_plot_noyname) +
    plot_layout(ncol = 3, guides = "collect") &
    theme(legend.position = "bottom")

ggsave(paste0("figures/figures_simulated_mm_RA_family/meanPrecisionRecallF1score_tool_byAge.pdf"),
      device="pdf", height=11, width=7)
```



In [ ]: